



STA201: Elements of Statistics & Probability

R-Language Bonus Assignment

Use built-in R functions to solve the following problems-

Marks-3.0

Q1. Create the following vector-

90, 178, 547, 453, 189, 377, 264, 333, 289, 391, 320, 300, 210, 310, 121, 154, 248, 292, 368, 423

- (i) Find mean, median, standard deviation of the above values (from question 1)
- (ii) Find square root of all values
- (iii) Sort the values
- (iv) Find the length of the vector
- (v) Find summation and product of all values

Q2. Create the following data frame-

Number of Rooms	kwh
12	9
9	7
14	10
6	5
10	8

- (i) Find Pearson's correlation coefficient between two variables (Number of rooms & KWH)
- (ii) Estimate the regression coefficients of a regression of KWH on Number of rooms.
- (iii) Plot a scatter diagram of KWH against Number of rooms

Submission Guideline: Submit the code and output, as well as the cover page, as in the previous submission. If you submit, this will be counted as a fixed/bonus three marks (i.e., average marks of assignments, 30 will be converted to 12; so, 12+3=15). Take help from R- tutorials shared with you.